

Algorithmic Pseudocode Sample 11

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

Input: ϕ, θ , initial episodes K_{init} , total budget of episodes K_E ,
Init: $\phi' \leftarrow \phi, \theta' \leftarrow \theta, \mathcal{D} \leftarrow \emptyset$

- 1: **for** each initial episode $1, \dots, K_{\text{init}}$ **do**
- 2: Sample a batch $\mathcal{T}$ of M sequences using pre-trained policy π_θ
- 3: Score each sequence in $\mathcal{T}$
- 4: Add unique, valid sequences to replay memory $\mathcal{D}$
- 5: **end for**
- 6: **for** each episode $K_{\text{init}} + 1, \dots, K_E$ **do**
- 7: Sample a batch $\mathcal{T}$ of M sequences using current policy π_θ
- 8: Score each sequence in $\mathcal{T}$
- 9: Add unique, valid sequences to replay memory $\mathcal{D}$
- 10: $\phi \leftarrow \phi - \lambda_Q \hat{\nabla}_\phi J_Q(\phi|\mathcal{T})$ ▷ On-policy update of Q-function parameters
- 11: $\theta \leftarrow \theta - \lambda_\pi \hat{\nabla}_\theta J_\pi(\theta|\mathcal{T})$ ▷ On-policy update of policy parameters
- 12: $\alpha \leftarrow \alpha - \lambda_\alpha \hat{\nabla}_\alpha J_\alpha(\alpha|\mathcal{T})$ ▷ On-policy update of temperature
- 13: $\phi' \leftarrow \tau\phi' + (1 - \tau)\phi$ ▷ Update target parameters
- 14: $\theta' \leftarrow \tau\theta' + (1 - \tau)\theta$ ▷ Update average policy parameters
- 15: **for** each off-policy update **do**
- 16: $\phi \leftarrow \phi - \lambda_Q \hat{\nabla}_\phi J_Q(\phi|\mathcal{D})$
- 17: $\theta \leftarrow \theta - \lambda_\pi \hat{\nabla}_\theta J_\pi(\theta|\mathcal{D})$
- 18: $\alpha \leftarrow \alpha - \lambda_\alpha \hat{\nabla}_\alpha J_\alpha(\alpha|\mathcal{D})$
- 19: $\phi' \leftarrow \tau\phi' + (1 - \tau)\phi$
- 20: $\theta' \leftarrow \tau\theta' + (1 - \tau)\theta$
- 21: **end for**
- 22: **end for**
